

Immune and Lymphatic Health

Host defense, tolerance, inflammation, allergy, autoimmunity, lymphatic transport, immunologic measurement, and engineering

Volume 03H of the Health Atlas Research Suite

Purpose

Map immune and lymphatic health as regulated defense, tolerance, repair, transport and memory rather than as a simple strong-versus-weak axis.

Evidence, guidance, and standards cutoff: 13 August 2026

August 2026

Document status and intended use

This is a technical evidence synthesis and research-engineering reference, not individualized medical advice, a substitute for clinical evaluation, or authorization to use an experimental device on people. It distinguishes established knowledge from uncertain inference and frontier research. Diagnostic and therapeutic decisions require qualified professionals, appropriate reference standards, and applicable ethics and regulatory review.

Suite relationship

This volume is read with the Atlas of Human Health (master taxonomy), the Library Guide and Crosswalk, and the other domain volumes identified in the cross-reference chapter. Volume 03A, Cardiovascular Health and Measurement Engineering, remains the reference model for the suite.

Evidence boundary

This volume does not provide emergency allergy treatment instructions, immunosuppressive dosing, transplant protocols or interpretation of isolated autoantibodies. Immune measurements require syndrome, timing, treatment and specimen context.

How to use this volume

Immune health is context-dependent discrimination and control: preventing or clearing harmful agents, tolerating self and ordinary exposures, resolving inflammation, repairing tissue, sustaining vaccine memory, and trafficking cells and fluid without causing allergy, autoimmunity, chronic inflammation or immune paralysis. No single cytokine, antibody panel or “immune score” captures that capacity.

Questions this volume is designed to answer

1. What does a well-regulated immune system do across tissues and time?
2. How should defense, tolerance, inflammation, repair and lymphatic flow be measured?
3. Why are many immune biomarkers nonspecific?
4. Which endotypes are actionable and which remain research constructs?
5. How should immune assays and longitudinal sensors be validated without creating false certainty?

Scope

- Barrier, innate, adaptive, tolerance, memory, lymphatic and resolution biology.
- Immunodeficiency, autoimmunity, autoinflammation, allergy, transplant immunity, lymphedema and immune dysregulation.
- Cellular, functional, molecular, tissue, microbial and longitudinal measurement.
- Vaccination, replacement, immunomodulation, cellular/gene therapy and rehabilitation.
- Device/assay engineering, validity, biosafety, equity and data governance.

Reading conventions

- **Established:** replicated mechanisms, mature measurement standards, or interventions supported by convergent evidence.
- **Context-limited:** useful under defined populations, settings, reference methods, or intended uses; transportability must be demonstrated.
- **Frontier:** biologically plausible or technically promising but not yet validated for broad clinical claims.
- **Measurement claim:** always interpreted as an inference from a signal chain rather than a direct view of the latent biological state.

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Executive synthesis

Immune health is context-dependent discrimination and control: preventing or clearing harmful agents, tolerating self and ordinary exposures, resolving inflammation, repairing tissue, sustaining vaccine memory, and trafficking cells and fluid without causing allergy, autoimmunity, chronic inflammation or immune paralysis. No single cytokine, antibody panel or “immune score” captures that capacity.

Central organizing principle

Map immune and lymphatic health as regulated defense, tolerance, repair, transport and memory rather than as a simple strong-versus-weak axis.

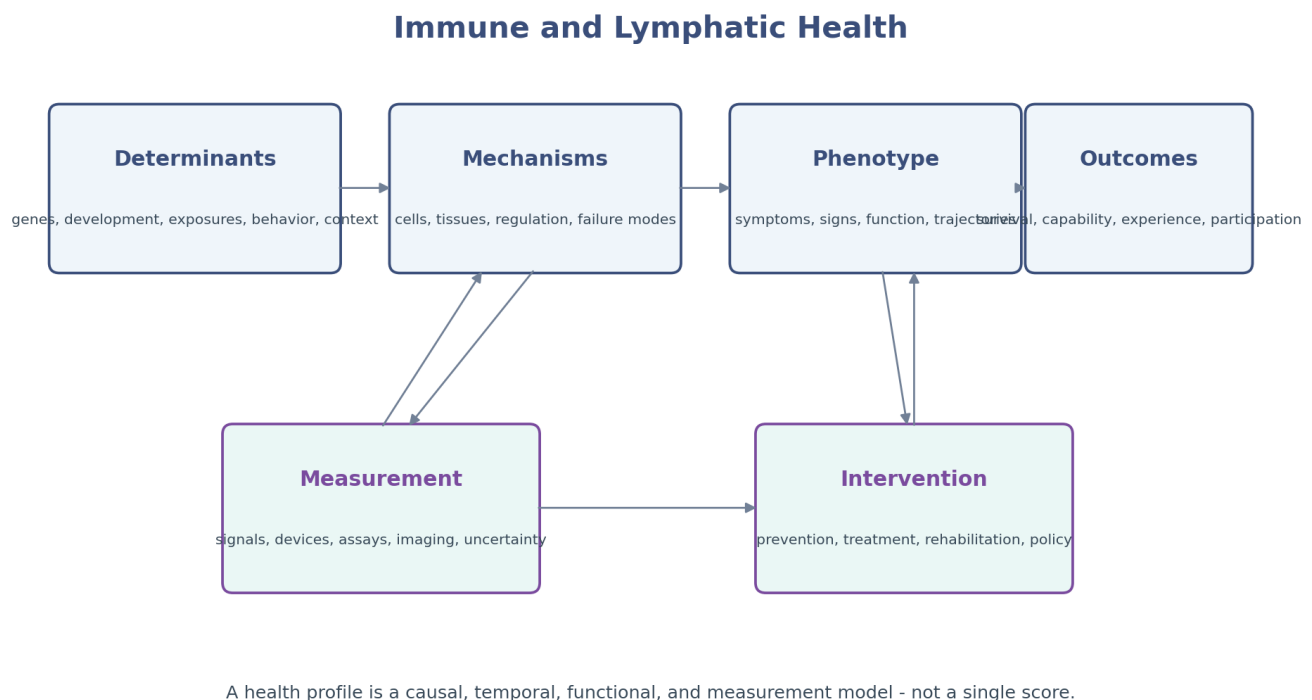


Figure 1.1: The causal and measurement architecture used throughout this volume.

1.1 What a complete profile requires

- A multiscale account from molecules and cells through tissues, organs, whole-person function, and context.
- A longitudinal account of baseline, reserve under challenge, variability, compensation, injury, recovery, and irreversible change.
- Separation of causal determinants from correlates, biomarkers, manifestations, and consequences.
- Explicit representation of measurement error, missingness, artifact, reference methods, population drift, and clinical actionability.

- Integration with other atlas domains rather than treating one specialty as a closed system.

Safety boundary

Anaphylaxis, severe infection, meningococcal-type illness, rapidly progressive weakness, airway swelling, sepsis, acute transplant dysfunction or severe immune-toxicity syndromes are emergencies. Experimental immune interventions and human specimen devices require biosafety, clinical oversight and long-term risk monitoring.

Historical development and changing models

The history of a domain is not decorative. It explains why present-day categories, instruments, thresholds, specialties, and blind spots look the way they do. Technical advances often changed the object of medicine itself: what could be seen, counted, localized, or treated became the preferred unit of explanation.

Period	Development	What it made possible	Residual limitation
Ancient to early modern	Accounts of contagion, immunity after illness and variolation preceded cellular mechanisms.	Established an empirical vocabulary for immunology while care remained descriptive.	Mechanisms were mixed with humoral, spiritual, and observational explanations.
Seventeenth to nineteenth centuries	Microscopy, lymphoid anatomy, germ theory and cellular pathology connected host and agent.	Connected anatomy and tissue structure to function and disease.	Static specimens could not explain regulation, kinetics, or lived function.
Late nineteenth to mid-twentieth century	Vaccines, serology, antibiotics, allergy testing, transplantation and immunosuppression made immune states actionable.	Made quantities, cell populations, and treatment response measurable.	Reference ranges and categories often hid heterogeneity and preanalytic error.
Late twentieth century	MHC, antibodies, receptors, cytokines, tolerance and monoclonal technologies enabled targeted explanations and therapies.	Linked genes, receptors, signaling pathways, and targeted therapy to phenotype.	Single-pathway explanations frequently underrepresented networks and context.
Twenty-first century	Single-cell/spatial omics, engineered immunity and systems immunology reveal heterogeneity and temporal states.	Enabled longitudinal, multimodal, high-dimensional, and precision approaches.	Clinical utility, transportability, cost, equity, and data governance remain limiting.
Current systems model	Immunology is now modeled as a coupled, adaptive system across molecules, cells, organs, behavior, environment, and care.	Supports cross-volume reasoning and explicit measurement models.	Education, reimbursement, datasets, and specialties still reinforce silos.

2.0.1 Accounts of contagion, immunity after illness and variolation preceded cellular mechanisms.

Scientific consequence. Established an empirical vocabulary for immunology while care remained descriptive.

What remained unresolved. Mechanisms were mixed with humoral, spiritual, and observational explanations.

Evidence anchors: ^[3,4]

2.0.2 Microscopy, lymphoid anatomy, germ theory and cellular pathology connected host and agent.

Scientific consequence. Connected anatomy and tissue structure to function and disease.

What remained unresolved. Static specimens could not explain regulation, kinetics, or lived function.

Evidence anchors: [3,4]

2.0.3 Vaccines, serology, antibiotics, allergy testing, transplantation and immunosuppression made immune states actionable.

Scientific consequence. Made quantities, cell populations, and treatment response measurable.

What remained unresolved. Reference ranges and categories often hid heterogeneity and preanalytic error.

Evidence anchors: [3,4]

2.0.4 MHC, antibodies, receptors, cytokines, tolerance and monoclonal technologies enabled targeted explanations and therapies.

Scientific consequence. Linked genes, receptors, signaling pathways, and targeted therapy to phenotype.

What remained unresolved. Single-pathway explanations frequently underrepresented networks and context.

Evidence anchors: [3,4]

2.0.5 Single-cell/spatial omics, engineered immunity and systems immunology reveal heterogeneity and temporal states.

Scientific consequence. Enabled longitudinal, multimodal, high-dimensional, and precision approaches.

What remained unresolved. Clinical utility, transportability, cost, equity, and data governance remain limiting.

Evidence anchors: [3,4]

2.0.6 Immunology is now modeled as a coupled, adaptive system across molecules, cells, organs, behavior, environment, and care.

Scientific consequence. Supports cross-volume reasoning and explicit measurement models.

What remained unresolved. Education, reimbursement, datasets, and specialties still reinforce silos.

Evidence anchors: [1,18]

Normal foundations: structure, physiology, and regulation

Normal physiology is represented here as regulated ranges and adaptive control, not as one fixed number. Many variables are intentionally dynamic, tissue-specific, state-dependent, and coupled to other systems. Disease can arise from primary damage, maladaptive compensation, loss of reserve, mistimed regulation, or mismatch between organism and environment.

3.0.1 Barrier immunity and tissue sentinels

Core principle. Skin, mucosa, secretions, resident phagocytes and epithelial sensing prevent invasion while tolerating ordinary exposures.

Control architecture. Tight junctions, antimicrobial peptides, mucus, cilia, pH, microbiota, pattern-recognition receptors and local cytokines.

Failure modes. Barrier breakdown, dysbiosis, excessive inflammation, allergy, infection or chronic tissue remodeling.

Observable consequences. Barrier function, microbiology, local cytokines, imaging, symptoms and challenge responses.

Cross-system coupling. Skin, respiratory, GI, reproductive, neural and environmental systems.

Evidence anchors: [3,4,9]

3.0.2 Innate recognition and effector response

Core principle. Germline-encoded sensors trigger phagocytosis, inflammation, complement, interferons and cell death programs.

Control architecture. Pattern/damage recognition, inflammasomes, complement regulation, cytokine networks and metabolic state.

Failure modes. Immune deficiency, sepsis, autoinflammation, complement injury or collateral tissue damage.

Observable consequences. Cell counts/function, cytokines, complement, microbial burden, fever and organ responses.

Cross-system coupling. Infection, coagulation, metabolism, endothelium and nervous system.

Evidence anchors: [3,4,12]

3.0.3 Antigen presentation and adaptive immunity

Core principle. Antigen-specific T and B lymphocytes expand, differentiate, form memory and coordinate cellular or antibody responses.

Control architecture. MHC presentation, co-stimulation, cytokines, germinal centers, receptor diversity and regulatory cells.

Failure modes. Immunodeficiency, autoimmunity, allergy, rejection, immune escape or exhaustion.

Observable consequences. Flow cytometry, antibody specificity/titer, functional assays, repertoire and clinical protection.

Cross-system coupling. Vaccination, cancer, transplantation, pregnancy and microbiome.

Evidence anchors: [3,8,13]

3.0.4 Tolerance and self-nonsel regulation

Core principle. Central and peripheral checkpoints prevent destructive responses to self while preserving defense.

Control architecture. Deletion, anergy, regulatory cells, immune checkpoints, tissue privilege and tolerogenic context.

Failure modes. Autoimmunity, autoinflammation, allergy or excessive immunosuppression.

Observable consequences. Autoantibodies, complement, cellular phenotype, tissue pathology and longitudinal activity.

Cross-system coupling. Endocrine, renal, musculoskeletal, nervous, skin and reproductive systems.

Evidence anchors: [3,4,6]

3.0.5 Immunologic memory and vaccination

Core principle. Memory cells and durable antibody-producing cells accelerate future protection after infection or immunization.

Control architecture. Antigen design, dose, route, adjuvant, germinal-center maturation, age and immune state.

Failure modes. Waning protection, poor response, immune imprinting, adverse events or inequitable coverage.

Observable consequences. Clinical effectiveness, neutralization/functional assays, memory phenotype and breakthrough severity.

Cross-system coupling. Public health, infection, aging, maternal and child health.

Evidence anchors: [3,8,33]

3.0.6 Allergy and hypersensitivity

Core principle. Protective immune effectors can be misdirected toward otherwise harmless antigens or host tissues.

Control architecture. IgE/mast cells, immune complexes, antibodies, T cells, epithelial alarmins and barrier defects.

Failure modes. Anaphylaxis, asthma, rhinitis, dermatitis, food/drug allergy and delayed reactions.

Observable consequences. History, challenge where safe, skin/specific-IgE tests, tryptase and organ-specific measures.

Cross-system coupling. Respiratory, skin, GI, pharmacology and environment.

Evidence anchors: [7,14,15]

3.0.7 Lymphatic transport and immune organs

Core principle. Lymphatic vessels return interstitial fluid and traffic antigens/cells through nodes, spleen, thymus and mucosal tissues.

Control architecture. Pressure gradients, valves, muscle movement, lymphangiogenesis and organ microarchitecture.

Failure modes. Lymphedema, impaired surveillance, infection, protein loss or metastatic spread.

Observable consequences. Limb volume, bioimpedance, imaging, node/tissue pathology and infection outcomes.

Cross-system coupling. Cardiovascular, cancer, musculoskeletal, skin and rehabilitation.

Evidence anchors: [4,13,16]

3.0.8 Immune-metabolic and neuroimmune coupling

Core principle. Immune-cell function depends on metabolic programs and communicates bidirectionally with neural, endocrine and autonomic signals.

Control architecture. Glucose/lipid/amino-acid metabolism, hormones, sympathetic/vagal signaling, circadian timing and cytokines.

Failure modes. Chronic inflammation, sickness behavior, fatigue, cachexia, insulin resistance or dysautonomia.

Observable consequences. Metabolites, cytokines, cell states, autonomic measures, symptoms and function.

Cross-system coupling. Endocrine, brain/behavior, aging, exercise and cross-system pathology.

Evidence anchors: [21,22]

3.0.9 Resolution, repair and immune homeostasis

Core principle. Effective defense must transition to pathogen clearance, resolution, tissue repair and memory without persistent damage.

Control architecture. Pro-resolving mediators, efferocytosis, regulatory cells, matrix remodeling and removal of antigen.

Failure modes. Chronic inflammation, fibrosis, immune paralysis, recurrent infection or post-acute syndromes.

Observable consequences. Temporal immune profiles, tissue repair, organ function, symptoms and recurrence.

Cross-system coupling. All organ systems, rehabilitation and life course.

Evidence anchors: [4,12,16]

3.1 Useful quantitative relationships

These equations are compact models. Their variables require operational definitions, units, boundary conditions, and a statement of when the approximation fails.

Relationship	Expression	Interpretation	Limits
Bayesian test interpretation	$\text{Posterior odds} = \text{Prior odds} \times \text{Likelihood ratio}$	An antibody or immune test changes probability rather than creating a diagnosis by itself.	Requires applicable likelihood ratios and a defensible pretest probability.
Vaccine effectiveness	$VE = 1 - RR$	Relative reduction in an outcome among vaccinated compared with unvaccinated groups under the study design.	Confounding, waning, variants, outcome definition and absolute risk matter.
Lymph flow concept	$Q_L \approx (P_i - P_L)/R_L$	A simplified pressure-resistance representation of lymphatic transport.	Lymphatic pumping, valves, tissue compliance and nonlinear recruitment make this only conceptual.

Determinants, heterogeneity, and causal architecture

No universal percentage can divide health into genetics and environment. Contributions vary by outcome, age, population, exposure distribution, measurement scale, and causal estimand. Genetic liability can alter sensitivity to an exposure; exposures can alter gene regulation; both operate through development, behavior, social conditions, and access to prevention or care.

Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Genetic architecture and family history	Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within immune and lymphatic health.	Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.	Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.	Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.
Development and life course	Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.	Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.	Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.	Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.
Nutrition and metabolism	Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.	Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.	Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.	Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.

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Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Physical activity, conditioning, and sedentary behavior	Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.	Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.	Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.	Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.
Sleep, circadian timing, and recovery	Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.	Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.	History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.	Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.
Stress, emotion, and behavior	Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.	Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.	Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.	Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.
Social, economic, cultural, and care conditions	Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.	Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.	Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.	Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.
Environmental and occupational exposures	Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.	Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.	Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.	Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.

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Determinant	Main pathways	Leverage	Measurement	Interpretive caution
Health care, medication, and iatrogenic effects	Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.	Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.	Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.	Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

4.0.1 Genetic architecture and family history

Mechanistic pathways. Monogenic variants, structural variation, mitochondrial inheritance, polygenic liability, and pleiotropy can alter baseline structure, regulatory set points, reserve, treatment response, and vulnerability within immune and lymphatic health.

Potentially modifiable components. Genotype is not modifiable, but surveillance, exposure reduction, reproductive counseling, and targeted therapy may change outcomes.

How to measure it. Three-generation pedigree when relevant; phenotype-first testing; chromosomal, panel, exome/genome, repeat-expansion, mitochondrial, or pharmacogenomic assays selected by question.

Causal and equity cautions. Heritability is population- and environment-specific; ancestry is not a biological proxy for race; variants of uncertain significance must not be treated as diagnoses.

Evidence anchors: [5,18]

4.0.2 Development and life course

Mechanistic pathways. Organ formation, critical periods, pubertal transitions, pregnancy, cumulative damage, repair capacity, and aging alter both baseline function and response to later exposures.

Potentially modifiable components. Preconception health, maternal care, early nutrition, vaccination, education, injury prevention, and age-appropriate screening can shift trajectories.

How to measure it. Gestational and birth history, developmental milestones, longitudinal growth/function, reproductive history, age-related reserve and frailty.

Causal and equity cautions. Adult snapshots can misattribute developmental effects to current behavior; cohort and survivor effects distort age comparisons.

4.0.3 Nutrition and metabolism

Mechanistic pathways. Energy balance, protein and micronutrient sufficiency, dietary pattern, glycemic and lipid exposure, microbiome substrates, and toxic contaminants affect tissue maintenance and signaling.

Potentially modifiable components. Food environment, dietary pattern, supplementation for verified deficiency, treatment of metabolic disease, and reduction of harmful exposures.

How to measure it. Dietary history with known reporting error, anthropometry, relevant biomarkers, body composition, and repeated metabolic measures.

Causal and equity cautions. Single nutrients rarely capture the causal pattern; reverse causation and socioeconomic confounding are common.

Evidence anchors: ^[33]

4.0.4 Physical activity, conditioning, and sedentary behavior

Mechanistic pathways. Mechanical loading, perfusion, mitochondrial adaptation, autonomic regulation, insulin sensitivity, mood, sleep, and functional reserve.

Potentially modifiable components. Progressive activity and exercise adapted to disability, symptoms, safety, and goals; reduction of prolonged sedentary time.

How to measure it. Device-measured movement plus context, intensity, posture, strength, cardiorespiratory or task-specific capacity.

Causal and equity cautions. Wearables misclassify some activities and disability patterns; association with health partly reflects baseline capacity.

Evidence anchors: ^[33]

4.0.5 Sleep, circadian timing, and recovery

Mechanistic pathways. Hormonal timing, autonomic balance, immune regulation, memory, tissue repair, appetite, and exposure opportunity are synchronized by sleep and circadian systems.

Potentially modifiable components. Regular timing, treatment of sleep disorders, light management, work-schedule redesign, and reduction of disruptive substances when appropriate.

How to measure it. History, sleep diary, actigraphy, validated questionnaires, physiologic sleep testing when indicated.

Causal and equity cautions. Consumer sleep stages are model outputs, not direct polysomnographic equivalents; illness can cause poor sleep and vice versa.

Evidence anchors: ^[21]

4.0.6 Stress, emotion, and behavior

Mechanistic pathways. Threat appraisal, sympathetic and neuroendocrine activation, inflammation, coping behavior, adherence, substance use, and social interaction can alter exposure and physiology.

Potentially modifiable components. Safety, psychological treatment, social support, work redesign, coping skills, treatment of mental disorders, and reduction of chronic uncontrollable stressors.

How to measure it. Contextual history, validated reported measures, repeated physiology when useful, and assessment of material stressors rather than one generic stress score.

Causal and equity cautions. Biomarkers are nonspecific; blaming individuals for structural stress is scientifically and ethically wrong.

Evidence anchors: ^[21]

4.0.7 Social, economic, cultural, and care conditions

Mechanistic pathways. Income, education, housing, work, discrimination, language, legal status, relationships, culture, care access, and treatment burden shape exposures and options.

Potentially modifiable components. Policy, income and housing support, accessible care, trust, communication, transportation, accommodations, and community resources.

How to measure it. Person-reported context, geospatial and administrative data, access/quality measures, and participatory qualitative methods.

Causal and equity cautions. Do not convert social disadvantage into an immutable patient trait; structural causes require structural interventions.

4.0.8 Environmental and occupational exposures

Mechanistic pathways. Air, water, heat, noise, radiation, chemicals, pathogens, ergonomic load, shift work, and injury hazards act through dose, timing, route, mixtures, and vulnerability.

Potentially modifiable components. Source control, substitution, ventilation, personal protection, regulation, surveillance, and climate adaptation.

How to measure it. Job and location history, direct and modeled exposure, biomarkers where valid, and uncertainty in mixtures and cumulative dose.

Causal and equity cautions. Personal sensors may not estimate biologically effective dose; exposure misclassification usually attenuates but can also bias unpredictably.

Evidence anchors: ^[34,35]

4.0.9 Health care, medication, and iatrogenic effects

Mechanistic pathways. Access, diagnostic delay, treatment quality, polypharmacy, procedures, device complications, antimicrobial exposure, and discontinuity can prevent or cause harm.

Potentially modifiable components. Medication reconciliation, deprescribing when appropriate, safer systems, shared decisions, guideline adaptation, and follow-up.

How to measure it. Longitudinal medication/procedure record, adverse events, care transitions, adherence, treatment burden, and patient goals.

Causal and equity cautions. Confounding by indication makes treated people appear sicker; absence of care data can be mistaken for absence of disease.

Evidence anchors: ^[29]

Major disorders, failure modes, and clinically important phenotypes

The profiles below are not a replacement for a diagnostic manual. They are organized to connect mechanism, phenotype, measurement, intervention, and uncertainty. The primary unit is the immune or lymphatic disorder, response state, endotype, or functional phenotype, but cross-system processes and functional effects are retained.

5.0.1 Primary immunodeficiency / inborn errors of immunity

Definition and phenotype. Genetic defects affecting immune development, signaling, effector function or regulation.

Pathophysiology. Specific pathway failures produce characteristic infection, inflammation, autoimmunity, allergy or malignancy patterns.

Determinants and modifiers. Genotype, age, exposure, vaccination, microbiome, access to specialty diagnosis and treatment.

Assessment. Patterned history, counts/subsets, immunoglobulins, functional tests, genetics and infection/tissue evaluation.

Intervention logic. Infection prevention, immunoglobulin/cytokine/targeted therapy, transplantation or gene therapy according to defect.

Known limits and open questions. Variable penetrance and expanding gene-disease links complicate interpretation.

Suite links. 08 infection; 14 precision therapy; 07 development.

Evidence anchors: ^[3,8,18]

5.0.2 Acquired immunodeficiency

Definition and phenotype. Reduced immune capacity from infection, malignancy, malnutrition, systemic illness, medication, transplantation or age.

Pathophysiology. Loss or suppression of immune cells/functions alters pathogen spectrum, vaccine response and cancer surveillance.

Determinants and modifiers. Exposure, treatment intensity, nutrition, comorbidity, age and care access.

Assessment. Cause-specific testing, subsets/immunoglobulins/function, infection screening and treatment timeline.

Intervention logic. Treat cause where possible; prophylaxis, vaccination, replacement or immune restoration matched to risk.

Known limits and open questions. Routine counts can appear normal despite functional impairment.

Suite links. 08 infection; 14 pharmacology; 12 systems.

Evidence anchors: [3,8,10]

5.0.3 Systemic autoimmune disease

Definition and phenotype. Loss of tolerance causes autoantibody, immune-complex, cellular and inflammatory injury across organs.

Pathophysiology. Genetic susceptibility and environmental triggers interact with adaptive/innate networks and organ-specific vulnerability.

Determinants and modifiers. Sex, ancestry, infection, smoking/UV/drugs, hormones, social stress and access.

Assessment. Clinical pattern, autoantibodies interpreted by pretest probability, complement, inflammation, tissue and organ testing.

Intervention logic. Risk-adapted immunomodulation plus organ protection, vaccination, rehabilitation and toxicity monitoring.

Known limits and open questions. Positive antibodies are not diagnoses; activity and damage measures are imperfect.

Suite links. 03F renal; 03I MSK; 03J skin; 03C nervous.

Evidence anchors: [4,6,18]

5.0.4 Organ-specific autoimmunity

Definition and phenotype. Immune injury targets one dominant tissue such as thyroid, islet, gut, liver, skin or neuromuscular junction.

Pathophysiology. Antibodies and/or autoreactive cells impair function or destroy cells with secondary systemic effects.

Determinants and modifiers. Genetics, infection, microbiome, hormones, exposures and stochastic immune events.

Assessment. Organ function, specific antibodies, imaging/biopsy where indicated and exclusion of mimics.

Intervention logic. Replace lost function and suppress/modulate immunity when evidence supports net benefit.

Known limits and open questions. Biomarkers may predict risk without defining timing or need for treatment.

Suite links. 03D endocrine; 03E GI/liver; 03C nervous; 03J skin.

Evidence anchors: ^[6]

5.0.5 Autoinflammatory syndromes

Definition and phenotype. Dysregulated innate immune signaling causes recurrent or persistent sterile inflammation, often without classic autoantibodies.

Pathophysiology. Inflammasome, cytokine or innate-signaling defects generate fever and tissue injury.

Determinants and modifiers. Genotype, temperature/stress triggers, infection and access to rare-disease expertise.

Assessment. Pattern, inflammatory markers during attacks, genetics and targeted exclusion of infection/malignancy.

Intervention logic. Cytokine-targeted and supportive therapy based on mechanism; prevent organ damage.

Known limits and open questions. Variants of uncertain significance and partial phenotypes are common.

Suite links. 04 inflammation; 07 genetics/life course; 14 targeted therapy.

Evidence anchors: ^[3,4,18]

5.0.6 Allergic rhinitis, asthma and atopic disease

Definition and phenotype. Barrier and type-2 immune phenotypes producing airway, skin or mucosal symptoms to allergens and nonspecific triggers.

Pathophysiology. Epithelial alarmins, IgE, mast cells, eosinophils and T-cell programs interact with barrier and environment.

Determinants and modifiers. Family history, early exposures, pollution, occupation, microbiome and social conditions.

Assessment. History, organ function, skin/specific IgE testing when actionable, eosinophils and controlled challenge in selected cases.

Intervention logic. Exposure reduction where feasible, organ-directed therapy, immunotherapy/biologics for selected phenotypes and action plans.

Known limits and open questions. Sensitization does not equal clinical allergy; endotypes overlap.

Suite links. 03B respiratory; 03J skin; 09 environment.

Evidence anchors: [7,14,15]

5.0.7 Food and drug allergy / anaphylaxis

Definition and phenotype. Reproducible immune-mediated reactions to foods or medicines; anaphylaxis is a rapid systemic reaction with airway, breathing or circulation risk.

Pathophysiology. IgE-mediated mast-cell activation is common in immediate reactions; other immune mechanisms produce delayed organ injury.

Determinants and modifiers. Exposure dose/route, cofactors, age, asthma, labels, health literacy and access to emergency treatment.

Assessment. Detailed chronology, targeted testing, supervised challenge when appropriate, tryptase in selected acute events and phenotype documentation.

Intervention logic. Avoid confirmed trigger, emergency plans and epinephrine for anaphylaxis; delabel false drug allergy; desensitization/immunotherapy selectively.

Known limits and open questions. Tests have imperfect positive predictive value and labels cause major downstream harm.

Suite links. 14 pharmacology; 03B respiratory; 12 safety systems.

Evidence anchors: [7,10,29]

5.0.8 Chronic inflammatory arthritis and connective-tissue disease

Definition and phenotype. Immune-mediated synovial, vascular, fascial or connective-tissue inflammation causing pain, damage and systemic complications.

Pathophysiology. Adaptive and innate pathways sustain synovitis/vasculitis, autoantibodies and tissue remodeling.

Determinants and modifiers. Genetics, smoking, infection, sex/hormones, obesity, occupation and treatment access.

Assessment. Patterned history/exam, inflammatory markers, autoantibodies, ultrasound/MRI, organ screening and function.

Intervention logic. Early treat-to-target immunomodulation, exercise/rehabilitation, risk-factor management and surgery when damage requires.

Known limits and open questions. Symptoms, inflammation and structural damage can diverge; comparative sequencing evidence is incomplete.

Suite links. 03I MSK; 10 rehab; 14 therapeutics.

Evidence anchors: ^[6]

5.0.9 Vasculitis

Definition and phenotype. Inflammation of vessel walls causing ischemia, hemorrhage, aneurysm and organ-specific syndromes.

Pathophysiology. Autoantibodies, immune complexes, granulomatous or cell-mediated mechanisms vary by vessel size and disease.

Determinants and modifiers. Age, infection, drugs, immune disease and genetic background.

Assessment. Urgent organ assessment, inflammatory/autoimmune tests, imaging and biopsy when feasible without delaying treatment.

Intervention logic. Induction and maintenance immunosuppression matched to threat; infection prevention and vascular/organ support.

Known limits and open questions. Classification criteria are not diagnostic rules; biopsy and biomarkers can be negative.

Suite links. 03A vascular; 03F renal; 03C neurologic; 03J skin.

Evidence anchors: ^[4,6]

5.0.10 Transplant rejection and graft-versus-host disease

Definition and phenotype. Immune recognition of donor/recipient antigens injures transplanted organs or host tissues.

Pathophysiology. T cells, antibodies, innate activation and tissue injury interact with immunosuppression, infection and tolerance.

Determinants and modifiers. HLA/antibody compatibility, ischemia, adherence, immune state, microbiome and drug exposure.

Assessment. Organ function, donor-specific antibodies, drug levels, imaging/biopsy and infection evaluation.

Intervention logic. Prevention and phenotype-specific immunosuppression, plasma/cellular strategies and supportive care.

Known limits and open questions. Biomarkers and noninvasive monitoring do not yet fully replace tissue/context.

Suite links. 14 therapeutics; 08 infection; organ-specific volumes.

Evidence anchors: [3,18,29]

5.0.11 Lymphedema

Definition and phenotype. Protein-rich interstitial fluid accumulation from congenital or acquired lymphatic transport failure.

Pathophysiology. Reduced drainage promotes swelling, adipose/fibrotic remodeling, skin change and recurrent infection.

Determinants and modifiers. Node surgery/radiation, malignancy, infection, obesity, immobility and inherited development.

Assessment. History/exam, serial standardized volume, bioimpedance or imaging when diagnosis/plan uncertain, and function/skin assessment.

Intervention logic. Compression, exercise, skin care, weight and infection management; selected surgery; lifelong self-management support.

Known limits and open questions. Early detection thresholds and long-term comparative device evidence vary.

Suite links. 03J skin; 03I MSK; 10 rehabilitation; 03A fluid.

Evidence anchors: [13,16,17]

5.0.12 Immune dysregulation in sepsis and critical illness

Definition and phenotype. Life-threatening organ dysfunction arises from dysregulated host response with simultaneous inflammation, endothelial/coagulation injury and immune suppression.

Pathophysiology. Pathogen and damage signals activate interacting immune, neural, endocrine, metabolic and vascular networks.

Determinants and modifiers. Pathogen, source, host reserve, age, comorbidity, treatment timing, AMR and health-system capacity.

Assessment. Clinical syndrome, cultures/molecular tests, organ function, lactate/hemodynamics and source imaging; no single immune biomarker is definitive.

Intervention logic. Rapid antimicrobials/source control, organ support and reassessment; selected adjuncts by evidence.

Known limits and open questions. Heterogeneity has defeated many one-target immune therapies; post-sepsis trajectories are undermeasured.

Suite links. 04 cross-system; 08 infection; 12 critical care.

Evidence anchors: [10,12,29]

5.0.13 Post-infectious and persistent immune-associated syndromes

Definition and phenotype. Persistent symptoms and organ dysfunction after acute infection may involve residual injury, viral persistence, autoimmunity, dysautonomia, microvascular or metabolic changes and deconditioning.

Pathophysiology. Multiple mechanisms can coexist and broad labels mix phenotypes.

Determinants and modifiers. Pathogen/severity, reinfection, baseline health, sex, social context, care access and measurement timing.

Assessment. Rule out urgent/treatable disease; symptom, function, organ, autonomic and exertional assessment guided by phenotype.

Intervention logic. Pacing or graded rehabilitation individualized to response, symptom-targeted treatment and treatment of defined organ disease; avoid unvalidated universal cures.

Known limits and open questions. Causal biomarkers and treatment-responsive subgroups remain incompletely established.

Suite links. 01 whole-person; 04 cross-system; 10 rehabilitation; 05 cognition/behavior.

Evidence anchors: [16,30]

5.0.14 Immune-mediated adverse effects of therapy

Definition and phenotype. Drugs, biologics, vaccines, cellular therapy and devices can provoke hypersensitivity, autoimmunity, cytokine toxicity or immune suppression.

Pathophysiology. Off-target recognition, checkpoint release, cytokine amplification, antigen exposure or immune-cell depletion.

Determinants and modifiers. Agent, dose, route, host history, interactions, microbiome and recognition delay.

Assessment. Temporal phenotype, organ testing, alternative causes and mechanism-specific markers where useful.

Intervention logic. Stop/modify agent, emergency treatment or immunomodulation according to severity; document precisely rather than use vague labels.

Known limits and open questions. Rare events and confounding by indication challenge causal attribution.

Suite links. 14 therapeutics; 13 pharmacovigilance; organ volumes.

Evidence anchors: ^[7,29]

Assessment, clinical reasoning, and longitudinal profiling

Assessment is a staged inference process. The right test depends on the pretest question, urgency, expected change in management, burden, and the consequences of false reassurance or false alarm. A complete profile combines history, examination, validated measurements, function, trajectories, and context.

Stage	Purpose	Components	Output	Common failure
1. Question and urgency	Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.	Chief concern and goals; red flags; prior probability; relevant context and current immune and lymphatic health risks.	Urgent action, focused differential, or a prespecified measurement plan.	Testing without a question creates incidental findings and false precision.
2. History and lived phenotype	Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.	Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.	A longitudinal problem representation with hypotheses and missing information.	Administrative codes and checklists can erase chronology, context, and patient priorities.
3. Examination and basic measurements	Test hypotheses using standardized observation and low-burden validated measures.	Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.	Localization, severity estimate, safety screen, and indication for further testing.	Nonstandard technique and unblinded interpretation create systematic error.
4. Targeted laboratory, imaging, and physiology	Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.	Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.	Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.	Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.
5. Function and challenge testing	Measure reserve, adaptation, and real-world impact not visible at rest.	Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.	Capacity, symptom coupling, recovery, and safety limits.	Challenge protocols may not match daily demands and can be unsafe without screening.

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Stage	Purpose	Components	Output	Common failure
6. Longitudinal integration	Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.	Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.	Trajectory, response, relapse, and need for escalation or de-escalation.	Regression to the mean and changing devices can masquerade as improvement or decline.

6.0.1 1. Question and urgency

Purpose. Define the decision, symptom, screening target, research hypothesis, and time horizon; identify emergencies first.

Core components. Chief concern and goals; red flags; prior probability; relevant context and current immune and lymphatic health risks.

Interpretation. Urgent action, focused differential, or a prespecified measurement plan.

Failure modes. Testing without a question creates incidental findings and false precision.

6.0.2 2. History and lived phenotype

Purpose. Characterize onset, pattern, triggers, severity, variability, function, treatment response, and personal meaning.

Core components. Symptoms, timeline, development, family history, exposures, medications, behavior, sleep, psychosocial and occupational context.

Interpretation. A longitudinal problem representation with hypotheses and missing information.

Failure modes. Administrative codes and checklists can erase chronology, context, and patient priorities.

Evidence anchors: ^[30]

6.0.3 3. Examination and basic measurements

Purpose. Test hypotheses using standardized observation and low-burden validated measures.

Core components. Relevant physical, neurologic, behavioral, functional, and vital-sign examination; quality-controlled basic tests.

Interpretation. Localization, severity estimate, safety screen, and indication for further testing.

Failure modes. Nonstandard technique and unblinded interpretation create systematic error.

Evidence anchors: ^[4]

6.0.4 4. Targeted laboratory, imaging, and physiology

Purpose. Confirm or refute a mechanism, stage severity, identify treatable causes, or establish a reference baseline.

Core components. Select tests by expected information gain; document protocol, reference range, preanalytic conditions, and contrast/radiation/invasive burden.

Interpretation. Mechanism-informed diagnosis, prognostic information, or a revised probability distribution.

Failure modes. Reference intervals are not universal decision thresholds; incidental abnormalities may not explain the phenotype.

6.0.5 5. Function and challenge testing

Purpose. Measure reserve, adaptation, and real-world impact not visible at rest.

Core components. Exercise, task, stress, provocation, simulated activity, ecological monitoring, and patient-reported function as appropriate.

Interpretation. Capacity, symptom coupling, recovery, and safety limits.

Failure modes. Challenge protocols may not match daily demands and can be unsafe without screening.

6.0.6 6. Longitudinal integration

Purpose. Distinguish stable trait, reversible state, episodic event, progression, treatment effect, and measurement drift.

Core components. Repeated measures with synchronized context, interventions, adherence, quality, and prespecified meaningful-change thresholds.

Interpretation. Trajectory, response, relapse, and need for escalation or de-escalation.

Failure modes. Regression to the mean and changing devices can masquerade as improvement or decline.

Measurement and device landscape

A device does not measure health directly. It senses a physical or chemical proxy, transforms it, samples it, rejects or retains artifact, and maps the result to a claim. The claim may concern detection, diagnosis, severity, risk, response, or function; each requires a different validation design.

Measurement is an inference chain



Validation must test every link, including population, setting, motion, missingness, drift, and downstream use.

Figure 7.1: Generic inference chain from latent biology to a clinical or research claim.

Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Clinical immune phenotype and infection history	Noninvasive	Structured pattern across infection type/site, inflammation, allergy, autoimmunity, growth, family and treatment response.	Pretest phenotype and test selection.	Often the highest-yield discriminator.	Recall, referral bias, nonspecific symptoms and fragmented records.
Blood counts, differentials and morphology	Minimally invasive sample	Automated counts plus smear/flow characterization.	Cell abundance, maturation and flags.	Accessible first-line screen.	Normal numbers do not guarantee function; stress and drugs alter counts.
Immunoglobulins and specific antibodies	Minimally invasive sample	Immunoassay, nephelometry, binding or neutralization tests.	Isotypes, antigen-specific response, autoantibodies and monoclonal patterns.	Scalable humoral assessment.	Binding may not equal protection or pathogenicity; passive antibody and assay variability.

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Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Flow cytometry and cellular phenotyping	Minimally invasive sample	Multiparameter optical single-cell analysis.	Subsets, activation, exhaustion, clonality and rare populations.	High-resolution cellular map.	Gating, specimen delay, treatment and interlaboratory variation.
Cellular functional assays	Minimally invasive sample	Ex vivo stimulation, proliferation, cytotoxicity, oxidative burst, cytokine or killing assays.	Specific immune capacities under challenge.	Mechanistically closer than counts.	Labor-intensive, variable, artificial stimulation and transport sensitivity.
Complement testing	Minimally invasive sample	Functional hemolytic or pathway assays and component immunoassays.	Consumption, deficiency and pathway activity.	Useful in defined immune and renal syndromes.	Handling, acute-phase effects and pathway complexity.
Allergy skin tests and specific IgE	Minimally invasive/in vivo or sample	Controlled skin exposure or serum immunoassay for sensitization.	Sensitization pattern and probability support.	Rapid and useful when tied to history.	False positives, medication/skin effects, anaphylaxis risk and non-IgE mechanisms.
Supervised exposure or provocation challenge	In vivo, potentially high risk	Controlled antigen, food, drug, airway or environmental exposure under monitoring.	Clinical reactivity, threshold and mechanism-linked response.	Closest reference for selected allergy questions.	Can provoke severe reaction; context and protocol influence result.
Autoantibody and inflammatory-marker panels	Minimally invasive sample	Immunoassay, immunofluorescence and chemistry.	Antibody specificity/titer, complement, acute-phase and organ injury clues.	Supports defined clinical patterns.	Low specificity when ordered indiscriminately; assay/platform differences.
Tissue biopsy and spatial pathology	Invasive	Histology, immunostaining, molecular and spatial cellular analysis.	Pattern, cell infiltrates, immune deposits, infection, malignancy and damage.	Direct tissue context.	Sampling, procedural harm, timing and treatment effects.
Lymphatic imaging and limb-fluid measures	Noninvasive to minimally invasive	Circumference/volume, bioimpedance, ultrasound, lymphoscintigraphy, fluorescent or MR lymphography.	Fluid burden, anatomy, transport and surgical planning.	Quantifies otherwise hidden transport failure.	Position/hydration effects, limited standardization and tracer/procedure burden.

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Modality	Invasiveness	Signal / sensor	Outputs	Strengths	Limitations
Microbiologic and pathogen-host testing	Minimally invasive to sample	Culture, antigen, nucleic acid, sequencing, serology and susceptibility.	Pathogen, burden, resistance and host response evidence.	Directly actionable when syndrome and sample are appropriate.	Colonization, contamination, timing, prior therapy and incidental detection.
Cytokine, proteomic and metabolomic profiling	Minimally invasive sample/research	Multiplex affinity, mass spectrometry and multi-omics.	High-dimensional immune state and candidate endotypes.	Discovery and mechanistic stratification.	Batch, timing, confounding, multiple testing and weak actionability.
Immune repertoire and single-cell sequencing	Minimally invasive to research	T/B receptor sequencing and transcriptomic/epigenomic single-cell assays.	Clonality, cell states, trajectories and antigen-response hypotheses.	Exceptional biological resolution.	Cost, sampling, computational choices, privacy and uncertain clinical meaning.
Wearable inflammation and recovery proxies	Noninvasive	Temperature, heart rate/variability, sleep, activity, sweat or interstitial biomarkers.	Change detection and symptom context, not direct immune diagnosis.	Longitudinal low-burden monitoring.	Nonspecific signals, device drift, behavior confounding and alert burden.

7.0.1 Clinical immune phenotype and infection history

Measured quantity and transducer. Structured pattern across infection type/site, inflammation, allergy, autoimmunity, growth, family and treatment response.

Typical outputs. Pretest phenotype and test selection.

Strengths. Often the highest-yield discriminator.

Limitations and artifacts. Recall, referral bias, nonspecific symptoms and fragmented records.

Reference and validation strategy. Standardized definitions, longitudinal verification and records/microbiology review.

Clinical/research maturity. Established.

Evidence anchors: ^[3]

7.0.2 Blood counts, differentials and morphology

Measured quantity and transducer. Automated counts plus smear/flow characterization.

Typical outputs. Cell abundance, maturation and flags.

Strengths. Accessible first-line screen.

Limitations and artifacts. Normal numbers do not guarantee function; stress and drugs alter counts.

Reference and validation strategy. Repeat/context, morphology and functional follow-up.

Clinical/research maturity. Established.

Evidence anchors: ^[4]

7.0.3 Immunoglobulins and specific antibodies

Measured quantity and transducer. Immunoassay, nephelometry, binding or neutralization tests.

Typical outputs. Isotypes, antigen-specific response, autoantibodies and monoclonal patterns.

Strengths. Scalable humoral assessment.

Limitations and artifacts. Binding may not equal protection or pathogenicity; passive antibody and assay variability.

Reference and validation strategy. Controls, functional assay where needed, pretest probability and longitudinal response.

Clinical/research maturity. Established with context limits.

Evidence anchors: ^[3,8]

7.0.4 Flow cytometry and cellular phenotyping

Measured quantity and transducer. Multiparameter optical single-cell analysis.

Typical outputs. Subsets, activation, exhaustion, clonality and rare populations.

Strengths. High-resolution cellular map.

Limitations and artifacts. Gating, specimen delay, treatment and interlaboratory variation.

Reference and validation strategy. Panel controls, harmonization, proficiency and clinical correlation.

Clinical/research maturity. Established specialist.

Evidence anchors: ^[3,13]

7.0.5 Cellular functional assays

Measured quantity and transducer. Ex vivo stimulation, proliferation, cytotoxicity, oxidative burst, cytokine or killing assays.

Typical outputs. Specific immune capacities under challenge.

Strengths. Mechanistically closer than counts.

Limitations and artifacts. Labor-intensive, variable, artificial stimulation and transport sensitivity.

Reference and validation strategy. Healthy/disease controls, reproducibility and outcome-linked interpretation.

Clinical/research maturity. Established for selected indications.

Evidence anchors: ^[3,25]

7.0.6 Complement testing

Measured quantity and transducer. Functional hemolytic or pathway assays and component immunoassays.

Typical outputs. Consumption, deficiency and pathway activity.

Strengths. Useful in defined immune and renal syndromes.

Limitations and artifacts. Handling, acute-phase effects and pathway complexity.

Reference and validation strategy. Strict preanalytics, method-specific ranges and mechanism correlation.

Clinical/research maturity. Established.

Evidence anchors: ^[3,6]

7.0.7 Allergy skin tests and specific IgE

Measured quantity and transducer. Controlled skin exposure or serum immunoassay for sensitization.

Typical outputs. Sensitization pattern and probability support.

Strengths. Rapid and useful when tied to history.

Limitations and artifacts. False positives, medication/skin effects, anaphylaxis risk and non-IgE mechanisms.

Reference and validation strategy. Standard extracts/controls, trained use and challenge correlation.

Clinical/research maturity. Established.

Evidence anchors: [7]

7.0.8 Supervised exposure or provocation challenge

Measured quantity and transducer. Controlled antigen, food, drug, airway or environmental exposure under monitoring.

Typical outputs. Clinical reactivity, threshold and mechanism-linked response.

Strengths. Closest reference for selected allergy questions.

Limitations and artifacts. Can provoke severe reaction; context and protocol influence result.

Reference and validation strategy. Specialist protocol, rescue readiness, objective endpoints and follow-up.

Clinical/research maturity. Established reference in selected cases.

Evidence anchors: [7,27]

7.0.9 Autoantibody and inflammatory-marker panels

Measured quantity and transducer. Immunoassay, immunofluorescence and chemistry.

Typical outputs. Antibody specificity/titer, complement, acute-phase and organ injury clues.

Strengths. Supports defined clinical patterns.

Limitations and artifacts. Low specificity when ordered indiscriminately; assay/platform differences.

Reference and validation strategy. Pretest-driven algorithms, confirmatory methods and longitudinal clinical endpoints.

Clinical/research maturity. Established but frequently misused.

Evidence anchors: [4,6]

7.0.10 Tissue biopsy and spatial pathology

Measured quantity and transducer. Histology, immunostaining, molecular and spatial cellular analysis.

Typical outputs. Pattern, cell infiltrates, immune deposits, infection, malignancy and damage.

Strengths. Direct tissue context.

Limitations and artifacts. Sampling, procedural harm, timing and treatment effects.

Reference and validation strategy. Adequate specimen, integrated pathology and clinical concordance.

Clinical/research maturity. Established; spatial omics emerging.

Evidence anchors: [4,13,19]

7.0.11 Lymphatic imaging and limb-fluid measures

Measured quantity and transducer. Circumference/volume, bioimpedance, ultrasound, lymphoscintigraphy, fluorescent or MR lymphography.

Typical outputs. Fluid burden, anatomy, transport and surgical planning.

Strengths. Quantifies otherwise hidden transport failure.

Limitations and artifacts. Position/hydration effects, limited standardization and tracer/procedure burden.

Reference and validation strategy. Repeatability, reference imaging, outcomes and stage-specific thresholds.

Clinical/research maturity. Established to advanced.

Evidence anchors: [16,23]

7.0.12 Microbiologic and pathogen-host testing

Measured quantity and transducer. Culture, antigen, nucleic acid, sequencing, serology and susceptibility.

Typical outputs. Pathogen, burden, resistance and host response evidence.

Strengths. Directly actionable when syndrome and sample are appropriate.

Limitations and artifacts. Colonization, contamination, timing, prior therapy and incidental detection.

Reference and validation strategy. Specimen quality, controls, clinical adjudication and stewardship outcomes.

Clinical/research maturity. Established to emerging.

Evidence anchors: [10,11]

7.0.13 Cytokine, proteomic and metabolomic profiling

Measured quantity and transducer. Multiplex affinity, mass spectrometry and multi-omics.

Typical outputs. High-dimensional immune state and candidate endotypes.

Strengths. Discovery and mechanistic stratification.

Limitations and artifacts. Batch, timing, confounding, multiple testing and weak actionability.

Reference and validation strategy. Locked assays, repeated states, external validation and intervention response.

Clinical/research maturity. Research/frontier.

Evidence anchors: ^[20]

7.0.14 Immune repertoire and single-cell sequencing

Measured quantity and transducer. T/B receptor sequencing and transcriptomic/epigenomic single-cell assays.

Typical outputs. Clonality, cell states, trajectories and antigen-response hypotheses.

Strengths. Exceptional biological resolution.

Limitations and artifacts. Cost, sampling, computational choices, privacy and uncertain clinical meaning.

Reference and validation strategy. Cross-platform standards, orthogonal function and prospective utility.

Clinical/research maturity. Advanced research; selected clinical use.

Evidence anchors: ^[19,20,28]

7.0.15 Wearable inflammation and recovery proxies

Measured quantity and transducer. Temperature, heart rate/variability, sleep, activity, sweat or interstitial biomarkers.

Typical outputs. Change detection and symptom context, not direct immune diagnosis.

Strengths. Longitudinal low-burden monitoring.

Limitations and artifacts. Nonspecific signals, device drift, behavior confounding and alert burden.

Reference and validation strategy. Raw-data retention, personalized baselines, reference sampling and utility trials.

Clinical/research maturity. Emerging.

Evidence anchors: ^[23]

Engineering architectures, signal quality, and validation

Buildable systems should begin with an explicit intended use, measurand, reference method, population, environment, and failure policy. Prototype plausibility is not clinical validity. Raw data, quality channels, timing, calibration state, firmware version, and provenance should be retained so that errors can be reconstructed rather than hidden by a polished score.

Subsystem / design issue	Requirements	Dominant risks	Verification
Intended use and measurand	Define the precise immune and lymphatic health quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.	A broad wellness claim can hide several incompatible clinical claims.	Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.
Sensor and transduction	Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.	A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.	Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.
Timing and synchronization	Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.	Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.	Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.
Analog, acquisition, and digital integrity	Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.	Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.	Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.
Calibration and traceability	Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.	Model calibration to one device lot, site, or operator may not transport.	Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.
Artifact and quality channels	Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.	Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.	Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

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Subsystem / design issue	Requirements	Dominant risks	Verification
Algorithms and uncertainty	Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.	Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.	Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.
Safety, usability, cybersecurity, and privacy	Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.	Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.	Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

8.0.1 Intended use and measurand

Design requirements. Define the precise immune and lymphatic health quantity, population, setting, user, decision, sampling interval, and acceptable uncertainty.

Failure modes. A broad wellness claim can hide several incompatible clinical claims.

Bench verification. Written claims matrix; trace each claim to design inputs, reference method, dataset, and acceptance criterion.

Human validation boundary. Do not enroll people until the measurement purpose and risk controls are reviewable.

Evidence anchors: [23,26]

8.0.2 Sensor and transduction

Design requirements. Characterize sensitivity, range, noise, bandwidth, cross-sensitivity, hysteresis, saturation, drift, temperature dependence, and biocompatible interface.

Failure modes. A plausible-looking waveform can arise from motion, pressure, ambient energy, crosstalk, contamination, or algorithmic smoothing.

Bench verification. Calibrated physical/chemical standards, simulators, controlled perturbations, and failure injection across operating range.

Human validation boundary. Bench testing should precede contact with participants.

Evidence anchors: [26]

8.0.3 Timing and synchronization

Design requirements. Use explicit clocks, timestamp provenance, drift characterization, event markers, and synchronization uncertainty for multimodal data.

Failure modes. Millisecond-to-minute drift can create false lead-lag physiology or destroy event alignment.

Bench verification. Long-duration clock tests, known pulse trains, dropout/restart tests, and end-to-end latency measurement.

Human validation boundary. Store both device time and host/reference time.

8.0.4 Analog, acquisition, and digital integrity

Design requirements. Document gain, filtering, ADC performance, anti-aliasing, resolution, dynamic range, saturation recovery, packet loss, and compression.

Failure modes. Hidden filters distort morphology and phase; quantization and clipping may be mislabeled as biology.

Bench verification. Frequency response, noise spectral density, step response, injected signals, checksum and data-loss testing.

Human validation boundary. Preserve raw or minimally processed channels where ethically and technically possible.

Evidence anchors: ^[25]

8.0.5 Calibration and traceability

Design requirements. Specify calibration hierarchy, reference uncertainty, intervals, field checks, lot/device effects, and recalibration triggers.

Failure modes. Model calibration to one device lot, site, or operator may not transport.

Bench verification. Traceable standards, multi-device studies, interoperator studies, environmental chambers, and blinded comparison.

Human validation boundary. Calibration data and version must travel with each observation.

Evidence anchors: ^[25]

8.0.6 Artifact and quality channels

Design requirements. Measure contact, motion, pressure, temperature, ambient conditions, specimen quality, and operator state rather than inferring quality only from the primary signal.

Failure modes. Artifact may correlate with disease, disability, skin tone, age, or setting and create biased missingness.

Bench verification. Artifact challenge set, subgroup performance, quality-controlled exclusion rules, and sensitivity analyses.

Human validation boundary. Quality rejection must not silently exclude populations.

8.0.7 Algorithms and uncertainty

Design requirements. Separate signal processing, feature extraction, model inference, confidence, calibration, abstention, and human review.

Failure modes. Data leakage, spectrum bias, shortcut learning, nonstationarity, overfitting, and automation bias.

Bench verification. Locked external validation, temporal/geographic transport tests, calibration curves, decision analysis, adversarial and failure testing.

Human validation boundary. A model should abstain when inputs are outside validated conditions.

Evidence anchors: ^[24]

8.0.8 Safety, usability, cybersecurity, and privacy

Design requirements. Risk management, electrical and mechanical safety, biocompatibility, human factors, secure update, least privilege, audit, consent, and data minimization.

Failure modes. Use error, unintended data disclosure, unsafe remote control, vulnerable update paths, and treatment delays.

Bench verification. Hazard analysis, formative/summative usability, penetration and abuse-case testing, incident response drills, and postmarket monitoring.

Human validation boundary. Research exemptions do not remove the ethical duty to minimize foreseeable harm.

Evidence anchors: ^[27]

Human-contact engineering rule

Do not connect a human participant to a mains-referenced oscilloscope, programmer, charger, or non-medical ground path. Bench characterization should use simulators and isolated fixtures. Human research requires documented electrical safety, risk management, consent, privacy controls, and appropriate oversight.

Prevention, treatment, rehabilitation, and system intervention

Interventions are classified by target and level: upstream determinants, exposure reduction, behavioral support, prevention, diagnosis-linked treatment, acute rescue, chronic management, rehabilitation, assistive technology, and palliative or supportive care. Net benefit depends on baseline risk, absolute effect, burden, interactions, access, adherence, and the outcomes valued by the person.

Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Vaccination and passive immunization	Prevention	Induce or provide pathogen-specific protection and reduce severe disease/transmission depending on product.	One of the highest-impact public-health interventions; effectiveness varies by pathogen, host and coverage.	Adverse events, waning, access, misinformation and cold-chain/system constraints.	Coverage, break-through severity, safety surveillance, equity and immune response in selected high-risk groups.
Barrier and exposure protection	Prevention	Improve ventilation, hygiene, protective equipment, skin/mucosal care and exposure control.	Strong when matched to transmission or barrier mechanism.	Burden, unequal feasibility and risk compensation.	Exposure/process measures, infection/allergy outcomes and acceptability.
Antimicrobial stewardship and infection treatment	Treatment/system	Deliver timely effective therapy while minimizing unnecessary exposure and resistance selection.	Strong for diagnosed/suspected infection with ongoing reassessment and source control.	Toxicity, C. difficile, resistance and diagnostic uncertainty.	Cultures/susceptibility, response, dose/organ function, duration and stewardship metrics.
Immunoglobulin and immune-factor replacement	Replacement	Provide missing antibodies, complement-related products or cytokine support in defined defects.	Can prevent severe infection or bleeding/inflammation in selected conditions.	Infusion reactions, thrombosis, supply, cost and product variability.	Clinical events, trough/function, adverse effects and dose need.

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Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Corticosteroids and conventional immunosuppression	Treatment	Broadly suppress immune activation, proliferation or mediator production.	Rapid and effective in many immune diseases but cumulative toxicity is substantial.	Infection, metabolic/bone/psychiatric toxicity, malignancy and adrenal suppression.	Disease activity/damage, infection, blood pressure/glucose/bone and taper response.
Targeted biologic and small-molecule immunomodulation	Treatment/prevention	Block cytokines, receptors, cell populations, co-stimulation or intracellular pathways.	Major benefit in defined autoimmune, allergic and inflammatory phenotypes.	Serious infection, reactivation, paradoxical effects, thrombosis, cost and uncertain sequencing.	Mechanism-specific screening, disease target, toxicity, vaccination and function.
Allergen immunotherapy and desensitization	Disease-modifying/specialist	Repeated controlled exposure alters immune response or temporarily raises reaction threshold.	Effective for selected aeroallergen/venom disease; drug/food protocols are context-specific.	Systemic reactions, adherence burden and loss of protection after interruption for some approaches.	Symptoms/exposure, reactions, rescue use and sustained benefit.
Emergency anaphylaxis treatment	Acute rescue	Intramuscular epinephrine supports airway, vascular tone and mediator control; adjuncts do not replace it.	Time-critical standard treatment.	Delay, dosing/use error, biphasic reaction and inadequate access.	Clinical response, repeated dosing, observation and device usability.
Cellular, transplant and gene therapy	Potentially curative/advanced	Replace or re-engineer immune/hematopoietic function.	Transformative for selected inborn errors and cancers.	Conditioning, graft disease, cytokine/neurotoxicity, infection, clonal risk, cost and inequity.	Immune reconstitution, molecular correction, disease control and long-term safety.
Lymphedema rehabilitation and assistive technology	Rehabilitation	Compression, movement, self-care, skin protection and selected surgery improve fluid control and participation.	Supports symptom and infection reduction; requires sustained individualized use.	Garment burden, skin injury, cost and limited access.	Volume, skin, infection, symptoms, activity and goals.
Tolerance and microbiome-directed strategies	Frontier	Induce antigen-specific tolerance or alter host-microbe interactions without broad suppression.	Mechanistically promising with established success only in selected contexts.	Unintended immune effects, ecological disruption, weak product standardization and hype.	Mechanism, microbial ecology, clinical outcomes and long-term off-target effects.

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Intervention	Level	Target / mechanism	Evidence profile	Risks / burdens	Monitoring
Shared decision, self-management and mental-health support	Whole-person	Match treatment burden and uncertainty to goals while supporting adherence, stigma, fatigue and coping.	Improves implementation and patient-centered outcomes across chronic immune disease.	Access, fragmented services and digital exclusion.	Goals, function, symptoms, treatment burden, distress and participation.

9.0.1 Vaccination and passive immunization

Target and mechanism. Induce or provide pathogen-specific protection and reduce severe disease/transmission depending on product.

Evidence and expected outcome. One of the highest-impact public-health interventions; effectiveness varies by pathogen, host and coverage.

Risks, burden, and equity. Adverse events, waning, access, misinformation and cold-chain/system constraints.

Response monitoring. Coverage, breakthrough severity, safety surveillance, equity and immune response in selected high-risk groups.

Evidence anchors: [8,33]

9.0.2 Barrier and exposure protection

Target and mechanism. Improve ventilation, hygiene, protective equipment, skin/mucosal care and exposure control.

Evidence and expected outcome. Strong when matched to transmission or barrier mechanism.

Risks, burden, and equity. Burden, unequal feasibility and risk compensation.

Response monitoring. Exposure/process measures, infection/allergy outcomes and acceptability.

Evidence anchors: [9,34,35]

9.0.3 Antimicrobial stewardship and infection treatment

Target and mechanism. Deliver timely effective therapy while minimizing unnecessary exposure and resistance selection.

Evidence and expected outcome. Strong for diagnosed/suspected infection with ongoing reassessment and source control.

Risks, burden, and equity. Toxicity, *C. difficile*, resistance and diagnostic uncertainty.

Response monitoring. Cultures/susceptibility, response, dose/organ function, duration and stewardship metrics.

Evidence anchors: [10,11]

9.0.4 Immunoglobulin and immune-factor replacement

Target and mechanism. Provide missing antibodies, complement-related products or cytokine support in defined defects.

Evidence and expected outcome. Can prevent severe infection or bleeding/inflammation in selected conditions.

Risks, burden, and equity. Infusion reactions, thrombosis, supply, cost and product variability.

Response monitoring. Clinical events, trough/function, adverse effects and dose need.

Evidence anchors: [3]

9.0.5 Corticosteroids and conventional immunosuppression

Target and mechanism. Broadly suppress immune activation, proliferation or mediator production.

Evidence and expected outcome. Rapid and effective in many immune diseases but cumulative toxicity is substantial.

Risks, burden, and equity. Infection, metabolic/bone/psychiatric toxicity, malignancy and adrenal suppression.

Response monitoring. Disease activity/damage, infection, blood pressure/glucose/bone and taper response.

Evidence anchors: [6]

9.0.6 Targeted biologic and small-molecule immunomodulation

Target and mechanism. Block cytokines, receptors, cell populations, co-stimulation or intracellular pathways.

Evidence and expected outcome. Major benefit in defined autoimmune, allergic and inflammatory phenotypes.

Risks, burden, and equity. Serious infection, reactivation, paradoxical effects, thrombosis, cost and uncertain sequencing.

Response monitoring. Mechanism-specific screening, disease target, toxicity, vaccination and function.

Evidence anchors: [6,7,18]

9.0.7 Allergen immunotherapy and desensitization

Target and mechanism. Repeated controlled exposure alters immune response or temporarily raises reaction threshold.

Evidence and expected outcome. Effective for selected aeroallergen/venom disease; drug/food protocols are context-specific.

Risks, burden, and equity. Systemic reactions, adherence burden and loss of protection after interruption for some approaches.

Response monitoring. Symptoms/exposure, reactions, rescue use and sustained benefit.

Evidence anchors: ^[7]

9.0.8 Emergency anaphylaxis treatment

Target and mechanism. Intramuscular epinephrine supports airway, vascular tone and mediator control; adjuncts do not replace it.

Evidence and expected outcome. Time-critical standard treatment.

Risks, burden, and equity. Delay, dosing/use error, biphasic reaction and inadequate access.

Response monitoring. Clinical response, repeated dosing, observation and device usability.

Evidence anchors: ^[7,29]

9.0.9 Cellular, transplant and gene therapy

Target and mechanism. Replace or re-engineer immune/hematopoietic function.

Evidence and expected outcome. Transformative for selected inborn errors and cancers.

Risks, burden, and equity. Conditioning, graft disease, cytokine/neurotoxicity, infection, clonal risk, cost and inequity.

Response monitoring. Immune reconstitution, molecular correction, disease control and long-term safety.

Evidence anchors: ^[13,18]

9.0.10 Lymphedema rehabilitation and assistive technology

Target and mechanism. Compression, movement, self-care, skin protection and selected surgery improve fluid control and participation.

Evidence and expected outcome. Supports symptom and infection reduction; requires sustained individualized use.

Risks, burden, and equity. Garment burden, skin injury, cost and limited access.

Response monitoring. Volume, skin, infection, symptoms, activity and goals.

Evidence anchors: [16,17]

9.0.11 Tolerance and microbiome-directed strategies

Target and mechanism. Induce antigen-specific tolerance or alter host-microbe interactions without broad suppression.

Evidence and expected outcome. Mechanistically promising with established success only in selected contexts.

Risks, burden, and equity. Unintended immune effects, ecological disruption, weak product standardization and hype.

Response monitoring. Mechanism, microbial ecology, clinical outcomes and long-term off-target effects.

Evidence anchors: [3,9]

9.0.12 Shared decision, self-management and mental-health support

Target and mechanism. Match treatment burden and uncertainty to goals while supporting adherence, stigma, fatigue and coping.

Evidence and expected outcome. Improves implementation and patient-centered outcomes across chronic immune disease.

Risks, burden, and equity. Access, fragmented services and digital exclusion.

Response monitoring. Goals, function, symptoms, treatment burden, distress and participation.

Evidence anchors: [30,31,32]

What is known, not known, and frequently overstated

Uncertainty is not a single residual. It can arise from biological variability, measurement error, selection, confounding, limited follow-up, inadequate reference standards, endpoint substitution, poor external validity, causal ambiguity, or value disagreement. The correct response is to identify the uncertainty and design the next measurement or study.

Question	What is established	What remains uncertain	Why unresolved	Decisive next step
Can immune aging be reversed safely?	Aging changes cell composition, repertoire, inflammation and vaccine response.	Which interventions improve net host defense without autoimmunity or cancer risk.	Markers often shift without demonstrated clinical benefit.	Randomized studies with infection, vaccine, cancer, autoimmunity and function outcomes.
Can immune endotypes guide routine treatment?	Molecular and cellular clusters exist in many diseases.	Stability, causal meaning and incremental treatment benefit.	Clusters are platform- and cohort-sensitive and may reflect treatment or severity.	Prospective treatment-interaction trials using locked assays.
How much risk is genetic versus environmental?	Both inherited variation and environments contribute, often through interaction and mediation across the life course.	No single percentage applies to all outcomes, populations, ages, or definitions within immune and lymphatic health.	Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.	Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.
Can one composite score summarize health?	Composite scores can rank a defined outcome in a defined population and may improve workflow.	Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.	Different variables compensate, interact, and have different treatment implications; weights drift over time.	Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.

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Question	What is established	What remains uncertain	Why unresolved	Decisive next step
When is earlier detection beneficial?	Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.	Net benefit for many incidental, molecular, wearable, and multi-cancer signals.	Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.	Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.
Do digital biomarkers generalize?	Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.	Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.	Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.	Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.
What counts as meaningful change?	Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.	Thresholds for many biomarkers, sensors, and multidomain profiles.	Baseline severity, goals, adaptation, and treatment burden alter value.	Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

10.0.1 Can immune aging be reversed safely?

Established knowledge. Aging changes cell composition, repertoire, inflammation and vaccine response.

Unresolved issue. Which interventions improve net host defense without autoimmunity or cancer risk.

Why current evidence is insufficient. Markers often shift without demonstrated clinical benefit.

Research that would change confidence. Randomized studies with infection, vaccine, cancer, autoimmunity and function outcomes.

Evidence anchors: [8,22]

10.0.2 Can immune endotypes guide routine treatment?

Established knowledge. Molecular and cellular clusters exist in many diseases.

Unresolved issue. Stability, causal meaning and incremental treatment benefit.

Why current evidence is insufficient. Clusters are platform- and cohort-sensitive and may reflect treatment or severity.

Research that would change confidence. Prospective treatment-interaction trials using locked assays.

Evidence anchors: ^[20]

10.0.3 How much risk is genetic versus environmental?

Established knowledge. Both inherited variation and environments contribute, often through interaction and mediation across the life course.

Unresolved issue. No single percentage applies to all outcomes, populations, ages, or definitions within immune and lymphatic health.

Why current evidence is insufficient. Heritability changes with exposure distribution; family studies mix genes and shared environment; causal pathways overlap.

Research that would change confidence. Large diverse longitudinal cohorts with family structure, repeated exposure measurement, molecular data, and explicit causal estimands.

Evidence anchors: ^[18]

10.0.4 Can one composite score summarize health?

Established knowledge. Composite scores can rank a defined outcome in a defined population and may improve workflow.

Unresolved issue. Whether a score preserves causal meaning, transportability, subgroup calibration, and patient-important outcomes outside development conditions.

Why current evidence is insufficient. Different variables compensate, interact, and have different treatment implications; weights drift over time.

Research that would change confidence. Externally validated transparent components, decision-curve analysis, recalibration, and reporting of disagreement among domains.

Evidence anchors: ^[30]

10.0.5 When is earlier detection beneficial?

Established knowledge. Early detection helps when there is a detectable preclinical phase, an effective earlier intervention, and acceptable harms.

Unresolved issue. Net benefit for many incidental, molecular, wearable, and multi-cancer signals.

Why current evidence is insufficient. Lead-time, length, overdiagnosis, cascade testing, anxiety, and treatment harms can dominate.

Research that would change confidence. Randomized or strong quasi-experimental studies with patient-important outcomes and explicit overdiagnosis analysis.

Evidence anchors: ^[13]

10.0.6 Do digital biomarkers generalize?

Established knowledge. Some digital measures reliably quantify narrow behaviors or signals under controlled conditions.

Unresolved issue. Long-term meaning across devices, firmware, disability, skin characteristics, cultures, settings, and disease severity.

Why current evidence is insufficient. Proprietary processing, dataset shift, artifact-dependent missingness, and weak reference labels.

Research that would change confidence. Open protocols, raw-data retention, standards, multisite external validation, and equity-centered failure analysis.

Evidence anchors: ^[23]

10.0.7 What counts as meaningful change?

Established knowledge. Statistical change, measurement error, minimal detectable change, and patient-important change are distinct.

Unresolved issue. Thresholds for many biomarkers, sensors, and multidomain profiles.

Why current evidence is insufficient. Baseline severity, goals, adaptation, and treatment burden alter value.

Research that would change confidence. Anchor-based longitudinal studies integrating function, symptoms, goals, adverse effects, and resource use.

Evidence anchors: ^[30]

Research and development roadmap

The roadmap prioritizes work that changes decisions or reduces uncertainty, not merely work that generates more variables. Each project should declare the causal or measurement target, comparator, reference, population, sampling frame, missing-data plan, subgroup analysis, and stopping criteria before data collection.

Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
0. Immune-state challenge atlas	How do protective and harmful immune responses differ under standardized perturbation?	Diverse longitudinal cohort with vaccination or controlled ex vivo challenge, tissue/blood sampling, symptoms, function and outcomes.	Reproducible dynamic signatures linked to clinical protection or toxicity.	Static clusters with no predictive or causal value.
0. Low-burden lymphedema detector	Can multimodal home measurements detect reversible fluid change before fibrosis?	Bench-calibrated bioimpedance/geometry/temperature detection with reference imaging and longitudinal intervention response.	Individual-level detection that improves treatment timing and patient outcomes.	Device drift, body-composition confounding or alarms without benefit.
1. Reference map	What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for immune and lymphatic health?	Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.	Reproducible distributions and state-dependent reference models with quantified uncertainty.	Convenience sample presented as universal normality.
2. Causal determinants	Which modifiable exposures cause clinically and functionally important outcomes, and in whom?	Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.	Convergent effect estimates linked to plausible mechanisms and actionable interventions.	Association-only result marketed as prevention guidance.
3. Measurement validity	Does the new device or biomarker measure the intended quantity under real use?	Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.	Prespecified error and failure limits met across intended-use conditions.	High average correlation with systematic individual error or unreported exclusions.

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Priority	Question	Minimum credible design	Success criterion	Stop / failure criterion
4. Clinical utility	Does using the measurement improve decisions or outcomes compared with current care?	Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.	Net benefit on patient-important outcomes with acceptable implementation burden.	Improved prediction without changed decisions or outcomes.
5. Longitudinal multimorbidity	How does the domain interact with other conditions, treatments, and functional trajectories?	Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.	Models that support coordinated care rather than isolated disease optimization.	Single-disease endpoint worsens overall burden or conflicts with goals.
6. Implementation and governance	Can effective interventions be delivered safely, equitably, affordably, and sustainably?	Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.	Sustained adoption with monitored outcomes and reduced inequity.	Pilot success depends on exceptional staffing or hidden externalized costs.

11.0.1 0. Immune-state challenge atlas

Question. How do protective and harmful immune responses differ under standardized perturbation?

Design. Diverse longitudinal cohort with vaccination or controlled ex vivo challenge, tissue/blood sampling, symptoms, function and outcomes.

Success criterion. Reproducible dynamic signatures linked to clinical protection or toxicity.

Failure or stop criterion. Static clusters with no predictive or causal value.

Evidence anchors: ^[19]

11.0.2 0. Low-burden lymphedema detector

Question. Can multimodal home measurements detect reversible fluid change before fibrosis?

Design. Bench-calibrated bioimpedance/geometry/temperature with reference imaging and longitudinal intervention response.

Success criterion. Individual-level detection that improves treatment timing and patient outcomes.

Failure or stop criterion. Device drift, body-composition confounding or alarms without benefit.

Evidence anchors: ^[16,23]

11.0.3 1. Reference map

Question. What are the normal ranges, state transitions, and reserve responses across age, sex, ancestry, disability, environment, and life stage for immune and lymphatic health?

Design. Prospective multi-site reference cohort; standardized protocols; repeated measures; raw data and quality channels; deliberate inclusion of underrepresented groups.

Success criterion. Reproducible distributions and state-dependent reference models with quantified uncertainty.

Failure or stop criterion. Convenience sample presented as universal normality.

11.0.4 2. Causal determinants

Question. Which modifiable exposures cause clinically and functionally important outcomes, and in whom?

Design. Triangulation across randomized interventions, natural experiments, longitudinal cohorts, negative controls, genetics where appropriate, and mechanistic studies.

Success criterion. Convergent effect estimates linked to plausible mechanisms and actionable interventions.

Failure or stop criterion. Association-only result marketed as prevention guidance.

11.0.5 3. Measurement validity

Question. Does the new device or biomarker measure the intended quantity under real use?

Design. Bench metrology followed by blinded method comparison, repeatability, subgroup, artifact, transport, and longitudinal responsiveness studies.

Success criterion. Prespecified error and failure limits met across intended-use conditions.

Failure or stop criterion. High average correlation with systematic individual error or unreported exclusions.

Evidence anchors: ^[26]

11.0.6 4. Clinical utility

Question. Does using the measurement improve decisions or outcomes compared with current care?

Design. Randomized or pragmatic comparative study measuring downstream actions, benefits, harms, burden, equity, and cost.

Success criterion. Net benefit on patient-important outcomes with acceptable implementation burden.

Failure or stop criterion. Improved prediction without changed decisions or outcomes.

Evidence anchors: ^[32]

11.0.7 5. Longitudinal multimorbidity

Question. How does the domain interact with other conditions, treatments, and functional trajectories?

Design. Person-centered longitudinal cohort or trial with common data model, medication and intervention timelines, function, participation, and competing risks.

Success criterion. Models that support coordinated care rather than isolated disease optimization.

Failure or stop criterion. Single-disease endpoint worsens overall burden or conflicts with goals.

11.0.8 6. Implementation and governance

Question. Can effective interventions be delivered safely, equitably, affordably, and sustainably?

Design. Hybrid effectiveness-implementation studies, participatory design, workflow and workforce analysis, privacy/security review, and policy evaluation.

Success criterion. Sustained adoption with monitored outcomes and reduced inequity.

Failure or stop criterion. Pilot success depends on exceptional staffing or hidden externalized costs.

Evidence anchors: ^[32]

Cross-volume integration

The health atlas is deliberately polyhierarchical. A phenomenon can have one primary home while also participating in multiple causal and functional systems. Cross-references below identify the most important companion volumes and the reason for the link.

Volume	Companion topic	Relationship	Use this link when...
00	Atlas of Human Health	Master taxonomy, history of classification, and multi-axial architecture.	locating the primary home of a concept or mapping classifications.
13	Measurement, Evidence, Data, and Informatics	Shared metrology, causal inference, validation, interoperability, and data governance.	designing a study, sensor, dataset, model, or evidence claim.
14	Interventions and Therapeutics	Cross-domain pharmacology, procedures, behavioral interventions, rehabilitation, implementation, and safety.	moving from measurement to action.
15	Ethics, Law, Rights, Equity, and Governance	Consent, privacy, fairness, rights, research oversight, and distribution of benefits and harms.	people, identifiable data, invasive procedures, or consequential algorithms are involved.
03G	Hematologic Health	Blood cells, marrow, complement, antibodies and immune malignancy are shared substrates.	cell number, function or clonal disease is central.
08	Infection, Immunity, Host-Microbe and Biosecurity	Pathogens, antimicrobial resistance, microbiomes, vaccination programs and outbreak control.	host defense must be interpreted with agent and population context.
04	Cross-System Pathological Processes	Inflammation, sepsis, fibrosis, cancer and post-acute syndromes cross organs.	immune activation becomes multisystem.
03I	Musculoskeletal Health	Inflammatory arthritis, connective-tissue disease and rehabilitation.	pain or disability may be inflammatory, structural or mixed.
09	Environmental and Occupational Health	Allergens, pollutants, workplace and ecological exposures shape immune phenotypes.	prevention depends on exposure control.

Cross-reference rule

Reference the primary home for the full mechanism or standard; restate only what is necessary for the local argument. When a claim spans domains, record the bridge explicitly rather than assigning the entire phenomenon to one specialty.

Minimum data schema for a longitudinal domain profile

Layer	Minimum fields	Critical caution
Identity and context	Participant identifier, consent, demographics relevant to the question, setting, language, access, and social/environmental context.	Avoid collecting identity variables without purpose; preserve privacy and provenance.
Time	Absolute timestamps, timezone, age/life stage, episode boundaries, event markers, medication and exposure timing.	Clock drift and ambiguous time zones can invalidate multimodal inference.
Raw signal / specimen	Unprocessed samples, units, sensor channel, specimen type, collection protocol, calibration and storage state.	Derived scores without raw or quality data are difficult to audit.
Quality	Contact, motion, saturation, missingness, assay controls, operator notes, confidence, out-of-range flags.	Quality must be a first-class variable, not only an exclusion after analysis.
Reference	Comparator method, instrument, version, observer protocol, laboratory traceability, adjudication.	A weak reference caps the validity of the new system.
Phenotype and function	Symptoms, signs, diagnoses, severity, capability, participation, patient-reported outcomes, goals.	Disease codes alone do not represent lived outcome.
Intervention and response	Dose, route, device settings, behavior support, adherence, co-interventions, adverse events, response window.	Confounding by indication and treatment changes must be modeled.
Governance	Protocol version, data-use agreement, access log, retention, deletion, model version, incident record.	Data stewardship is part of scientific validity.

Glossary

Term	Operational meaning in this volume
Tolerance	Mechanisms preventing harmful immune responses to self or innocuous antigens.
Immunogenicity	Capacity of an exposure or product to elicit an immune response; not synonymous with clinical protection.
Lymph	Interstitial fluid, proteins, lipids and cells transported through lymphatic vessels.
Endotype	A proposed subtype defined by a distinct biological mechanism; it requires validated measurement and clinical relevance.
Allostasis	Adaptive regulation that changes physiological activity to meet anticipated or actual demands; chronic or repeated activation can carry cost.
Biomarker	Defined characteristic measured as an indicator of normal biology, pathogenic process, exposure, or response; not automatically a surrogate endpoint.
Function	What a person can do in body functions, activities, and participation, interpreted in context.
Homeostasis	Regulation of variables within viable ranges through feedback, feedforward, redundancy, and adaptation.
Phenotype	Observable or measurable characteristics arising from biology, development, environment, behavior, context, and measurement conditions.
Risk factor	Characteristic associated with future outcome; causal status and modifiability must be established separately.
Calibration	Establishing the relationship between an instrument output and a reference quantity under specified conditions.
Clinical validity	Degree to which a measurement or model is associated with or predicts the clinical state or outcome it claims to represent.
Intended use	The specific purpose, population, setting, user, input, and decision for which a device or model is designed.
Reference method	Comparator selected to estimate truth for validation; every reference has its own error and scope.
Reserve	Capacity to maintain function under challenge beyond resting baseline.
Trajectory	Direction, rate, variability, and reversibility of change over time.

Selected references and current standards

The bibliography emphasizes official standards, current guidelines, landmark primary work, and major syntheses. Inclusion does not imply that every recommendation is universal or that institutional endorsement removes uncertainty. Current online materials were checked through the stated evidence cutoff.

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Version note

Foundation edition generated August 2026. Evidence, guidance, and standards cutoff: 13 August 2026. The suite should be versioned when major classifications, guidelines, device standards, or decisive evidence materially change the map.